Lab: Neural Correlates of Cognition

> **Learning Goal:** Map belief updating, attention, and cognitive control to neural circuits and signals.

Part 1: Attention Networks in Action

Scenarios: For each, identify whether dorsal (top-down) or ventral (bottom-up) attention is dominant, and explain why.

1. A lifeguard scanning a pool for struggling swimmers
2. A student startled by a fire alarm during an exam
3. A chess player analyzing the board position
4. A pedestrian hearing tires screech behind them

Write your response here

Part 2: P300 and Belief Updating

> **Learning Goal:** Understand ERP signatures of prediction error and belief updating.

Exercise: Design an experiment to produce a P300 response.

1. What repeated stimulus would you use?
2. What surprising stimulus would you insert?
3. Predict: Would the P300 be larger for rare or common deviants? Why?
4. How would the P300 change if you told participants in advance what the deviant would be?

Write your response here

Part 3: Computational Psychiatry Case Analysis

> **Learning Goal:** Apply the precision-weighting framework to clinical cases.

For each case, identify: (a) Which precision is disrupted? (b) Too high or too low? (c) What is the predicted neural mechanism?

1. A person with social anxiety interprets every neutral facial expression as threatening
2. A patient with ADHD cannot sustain focus on homework but can hyperfocus on a video game
3. A person with depression cannot feel pleasure from activities they used to enjoy (anhedonia)

Write your response here

Part 4: Working Memory and Free Energy

> **Learning Goal:** Connect PFC persistent activity to belief maintenance.

Thought experiment: You're holding a 7-digit phone number in mind while walking to write it down.

1. What brain region is maintaining this representation?
2. What happens if someone talks to you mid-walk (interference)?
3. How does Active Inference explain why the number "fades" if not rehearsed?
4. Why is working memory limited to ~4-7 items? Speculate using the accuracy-complexity trade-off.

Write your response here

Part 5: Reflection

In 150 words, reflect: Could computational psychiatry lead to personalized treatments — where a patient's specific precision profile is measured and targeted? What would be needed to make this a reality?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Network identification	Dorsal vs. ventral attention
2	Experiment design	P300 and belief updating
3	Clinical reasoning	Precision disruption in disorders
4	Mechanistic reasoning	PFC and working memory
5	Futures reflection	Computational psychiatry applications